

Semiconductor Fabrication Technology Roadmap

1. Advanced Process Nodes

High-volume manufacturing of the 3nm Gate-All-Around (GAA) nanosheet node commenced in Fab 7.

Nanosheet architecture provides a 15% performance enhancement at matched power compared to FinFET.

Wafer Yield & Defect Density

2. Defect Density Benchmarks

D0 defect density on commercial 300mm wafers reached 0.08 defects per square centimeter.

Wafer yield across high-density SRAM test vehicles exceeded 84.5%.